CHEMISTRY: Sample Practice Exam (KEY) **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

SECTION I — MULTIPLE CHOICE

CALCULATORS CANNOT BE USED IN THIS SECTION

Each question is worth one point: one point for the correct answer. Completely fill in one bubble per question.

Questions 1 – 3 refer to the following terms.

(A) Electron Affinity
(B) Shielding Effect
(C) Ionization Energy
(D) Electronegativity
(E) Atomic Radius

1) The energy change that occurs when a neutral atom in the gas phase gains an electron to form an anion.

1 ● B C D E

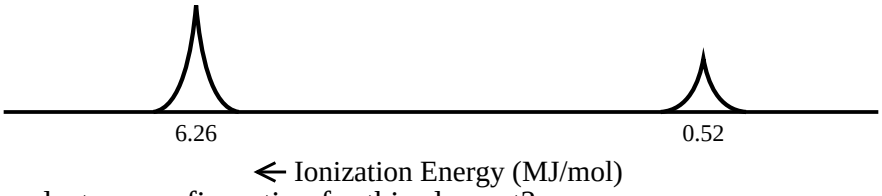
2) The reduction of the attractive force between the nucleus and the valence electrons, caused by the inner (core) electrons.

2 A ● C D E

3) One-half the distance between the nuclei of two identical atoms that are bonded together.

3 A B C D ●

Questions 4 and 5 refer to the photoelectron spectrum below.



4) What is the electron configuration for this element?

(A) $1s^2 2s^2$ (B) $1s^2 2s^1$ (C) $1s^2 2s^2 2p^1$ (D) $1s^2 2p^1$ (E) $1s^1 2s^2$

4 A ● C D E

5) Identify the element that would produce this spectrum.

(A) H (B) He (C) Li (D) Be (E) B

5 A B ● D E

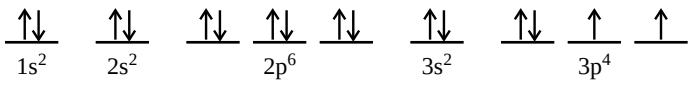
6) What mass of hydrogen gas is produced when 3.0 mol of magnesium react with excess hydrochloric acid according to the process outlined below?

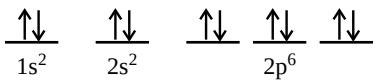
$$Mg + 2\ HCl \rightarrow MgCl_2 + H_2$$

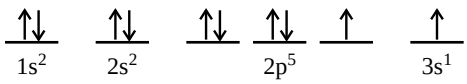
(A) 3.0 g H_2 (B) 6.0 g H_2 (C) 9.0 g H_2 (D) 12.0 g H_2 (E) 18.0 g H_2

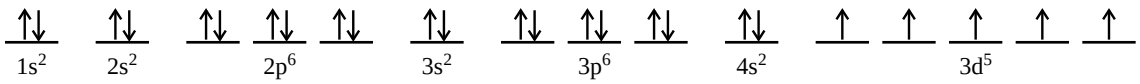
6 (A) (●) (C) (D) (E)

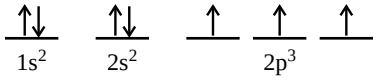
Questions 7 – 10 refer to the orbital diagrams below.

(A) 

(B) 

(C) 

(D) 

(E) 

7) Represents an atom with six valence electrons.

7 (●) (B) (C) (D) (E)

8) Does not violate any rules for filling orbitals and would not be attracted to a magnetic field.

8 (A) (●) (C) (D) (E)

9) Is in an excited state.

9 (A) (B) (●) (D) (E)

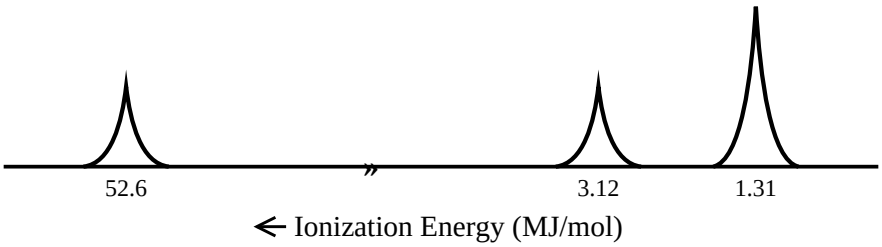
10) Represents a noble gas.

10 (A) (●) (C) (D) (E)

SECTION II — FREE RESPONSE

Show your work in the open area of each box; write only your final answer in the white FINAL ANSWER cell where provided. Point values for each part are printed beneath the question.

11) Use the photoelectron spectrum below to answer parts (a) – (d).



a. Identify the element that would produce the above spectrum.
One point for the correct answer.

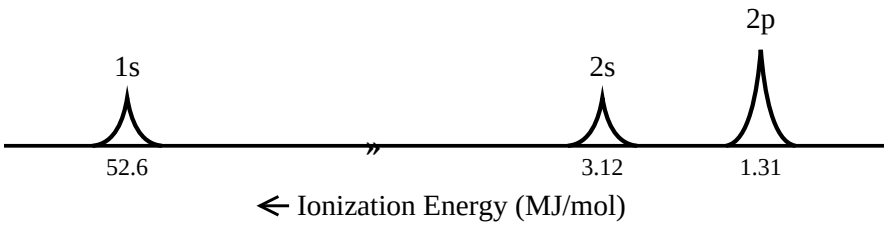
11a

Oxygen (O)

b. Label each peak on the spectrum with the numbers and letters that indicate its associated shell and subshell.
One point for the correct answer.

11b

LABEL THE SPECTRUM BELOW

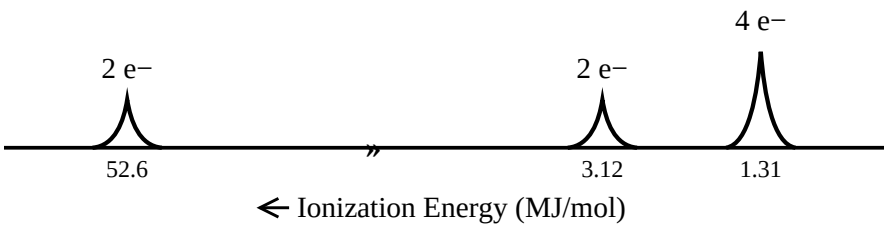


(Question 11, continued)

c. Indicate the number of electrons that are associated with each peak on the photoelectron spectrum.
One point for the correct answer.

11c

LABEL THE SPECTRUM BELOW



d. Identify the quantum level (e.g., n = 1, n = 2, …) that is holding the valence electrons in this element.
One point for the correct answer.

11d

n = 2

12) K^+ and S^{2-} are isoelectronic.

a. Which ion has the smaller radius?
One point for the correct answer.

12a

K^+

(Question 12, continued)

b. Explain why the radii of these two ions are different sizes.
One point for the correct explanation.

12b

K^+ and S^{2-} are isoelectronic: each ion has 18 electrons.

However, K^+ has 19 protons, while S^{2-} has only 16.

By Coulomb's law, the greater nuclear charge of K^+ attracts the same number of electrons more strongly, pulling them closer to the nucleus.

Therefore K^+ has the smaller ionic radius.

13) The following questions pertain to calcium.

a. Write the complete ground-state electron configuration for calcium, Ca.
One point for the correct answer.

13a

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

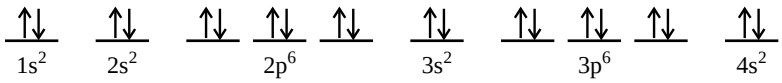
(Question 13, continued)

b. How many unpaired electrons does calcium have in its ground state? Justify your answer.

One point for the correct answer; one point for the justification. These points can be earned even if part (a) was incorrect, so long as your answer is consistent with the configuration you gave in part (a).

13b

JUSTIFY BELOW — ORBITAL DIAGRAM OR EXPLANATION

(Ca) 

Every sublevel through 4s is completely filled, so all of calcium's electrons are paired.
Calcium has zero unpaired electrons.

FINAL ANSWER NUMBER

0 (zero unpaired electrons)

c. Is a single gaseous calcium atom paramagnetic or diamagnetic? Justify your answer.

One point for the correct answer; one point for the justification.

13c

Calcium has no unpaired electrons: every sublevel in its ground-state configuration is completely filled, so the magnetic moments of the paired electrons cancel. With no net magnetic moment, a gaseous Ca atom is not attracted into a magnetic field.

FINAL ANSWER PARAMAGNETIC / DIAMAGNETIC

Diamagnetic